

Extension: LangChain, LangGraph

Tech stacks used in the demo code.

Outlines

- ✓ **LangChain:** The foundational toolkit for LLM chains and RAG.
- ✓ **LangGraph:** Graph-based orchestration for multi-agent workflows.

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LangChain

LangChain is a framework for developing applications powered by large language models.

Core Value Props:

- ✓ **Components:** Modular building blocks for LLM integration.
- ✓ **Off-the-shelf chains:** Built-in assemblages for high-level tasks.



LangChain: API Wrapping and HuggingFace Models



Unified interface for OpenAI, Gemini, Claude, NVIDIA NIM, and HuggingFace.

```
from langchain_openai import ChatOpenAI
llm = ChatOpenAI(model="gpt-5-mini",
temperature=0)

from langchain_google_genai import
ChatGoogleGenerativeAI
llm = ChatGoogleGenerativeAI(model="gemini-2.5-
flash")
```

LangChain : Simple RAG Tools

Indexing

Splitting &
Vector DB
Storage

Retrieval

Semantic search
for relevant
chunks

Generation

Augmenting
prompt with
retrieved
context

```
from langchain_text_splitters import (Chunking)
RecursiveCharacterTextSplitter
from langchain_chroma import Chroma (Vector DB)
```

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- ✓ **LangChain:** The foundational toolkit for LLM chains and RAG. (Part 1 and 2)
- ✓ **LangGraph:** Graph-based orchestration for multi-agent workflows. (Part 3)

LangGraph :

Orchestrate LLM workflows with cycles and complex branching logic.



Nodes: Agents or Functions



Edges: Connections (Control Flow)



State: Explicit shared memory

LangChain vs. LangGraph

Feature	LangChain	LangGraph
Structure	Linear Chains / DAGs	Cyclic Graphs (Loops)
State	Stateless	Fully Stateful (Shared Object)
Complexity	Simple Workflows / RAG	Multi-Agent / Agentic RAG
Use Cases	Chatbots, Q&A, Summarization	Self-Correction, HITL, Planning